

REQUIREMENTS ANALYSIS

FUNCTIONAL REQUIREMENTS:

1. Manage Bills and Customers

- Generate bills for customers.
- Assign bill numbers automatically
- Capture necessary customer information (name, contact, etc.)

2. Item Search

- Search medicines by brand name or barcode number.
- Display medicine details.

3. Sales

- Add multiple items to the bill.
- Update item quantities.
- Apply discounts(if applicable).
- Handle different payment methods.
- Automatically calculate subtotal, discount, net total, and taxes.
- Print the customer invoice with the bill number, date/time, item details, customer details, total amount, discount, and balance.

4. Stock Handling

- Display current stock.
- Alert the cashier if the stock is insufficient.
- Prevent the sale of expired medicine.

NON-FUNCTIONAL REQUIREMENTS:

1. **USABILITY:** Simple UI for easy use and quick transactions.
2. **PERFORMANCE:** The System must generate invoices quickly.

3. **SCALABILITY:** Should handle multiple users without affecting performance.
4. **RELIABILITY:** The System should not lose data in case of a power failure.
5. **SECURITY:** Customer data and billing records must be secure.

ITERATIONS:

1. Iteration 1: Sales Process

- Implement basic bill creation.
- Search/Select items.
- Add items to the cart.
- Calculate totals/discounts/taxes.
- Generate/Print invoices.
- Handle multiple payment methods.

2. Iteration 2: Customer Management

- Register customers by capturing their details.
- Store customers' past bills.
- Display customers' purchase history.

3. Iteration 3: Stock Handling

- Implement checks for stock availability.
- Update stock after each sale.
- Track expiry dates.

4. Iteration 4: Security

- Implement a login system for the cashier.
- Track cashier ID for each transaction.